



Review article

Early menopause is associated with increased risk of heart failure and atrial fibrillation: A systematic review and meta-analysis

Jiajun Liu^a, Xueshan Jin^b, Wanying Chen^a, Lan Wang^a, Ziyi Feng^a, Jieming Huang^{c,*}

^a First Clinical Medical College, Guangzhou University of Chinese Medicine, Guangzhou, China

^b School of Traditional Chinese Medicine, Jinan University, Guangzhou, China

^c The First Affiliated Hospital of Chinese Medicine, Guangzhou University of Chinese Medicine, Guangzhou, China

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ABSTRACT

Object: Menopause is linked to a higher risk of cardiovascular disease. However, it is unclear whether premature menopause (defined as menopause before the age of 40 years) or early menopause (defined as menopause before the age of 45 years) is associated with an increased risk of heart failure or atrial fibrillation. This study aimed to examine the most reliable evidence on the relationship between early menopause and the risk of heart failure and atrial fibrillation.

Methods: A comprehensive literature search was performed in three online databases, Embase, Web of Science, and PubMed, from database establishment to April 1, 2023. The results were presented as hazard ratios with 95 % confidence intervals. The I^2 statistic was employed to assess heterogeneity, and the Egger's test was used to determine publication bias.

Results: Nine cohort studies were included in the analysis, with a total of 6,255,783 postmenopausal women. Women with premature and early menopause had an increased risk of heart failure (HR: 1.39, 95 % CI: 1.31–1.47; HR: 1.23, 95 % CI: 1.10–1.37, respectively) and atrial fibrillation (HR: 1.15, 95 % CI: 1.01–1.31; HR: 1.08, 95 % CI: 1.04–1.13, respectively) when compared with women who had undergone menopause after the age of 45 years. Subgroup analysis showed that, compared with early menopause, premature menopause has a stronger association with an increased risk of heart failure and atrial fibrillation.

Conclusions: Women who undergo premature menopause or early menopause have a higher risk of heart failure and atrial fibrillation compared with women who undergo menopause in the normal age range. These reproductive factors need to be considered for measures that might reduce the risk of heart failure and atrial fibrillation.

1. Introduction

Menopause is the permanent cessation of the menstrual cycle, and is caused by reduced secretion of the ovarian hormones [1]. The average age of natural menopause ranges from 49 to 52 years, with slight differences between countries and regions [2,3]. Menopause before the age of 40, also known as premature menopause (PM, or primary ovarian insufficiency if menstruation ceased spontaneously), occurs in about 1 % of women [4]. Menopause before the age of 45, usually defined as early menopause (EM), affects approximately 10 % of women [5]. Heart failure (HF) and atrial fibrillation (AF) are global medical problems resulting in high morbidity and major health budget expenditures [6].

With the acceleration of population aging, the incidence of HF and AF is gradually increasing. The reported prevalence of HF is 0.1 %–6.7 %, affecting 26 million people [7]. The worldwide prevalence of AF has tripled over the past half century, and its prevalence in 2016 was approximately 5 % [8,9].

The decrease of ovarian estrogen secretion in postmenopausal women leads to varying degrees of vasomotor symptoms, also increases the risk of dementia, osteoporosis, and cardiovascular disease (CVD) events [10]. Among middle-aged and elderly women, CVD is the main cause of death. [8]. Previous literature has shown that early menopause is linked to a higher risk of CVD. A meta-analysis including 10 large cohort studies reported a significantly higher risk for CVD in women

* Corresponding author at: Women and Children's Center, The First Affiliated Hospital of Guangzhou University of Traditional Chinese Medicine, 16 Jichang Road, Baiyun District, 510045 Guangzhou, Guangdong, China.

E-mail address: 1223912678@qq.com (J. Huang).

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with primary ovarian insufficiency compared with women aged > 40 at menopause. [11]. However, the major endpoints were defined as ischemic heart disease and total cardiovascular events and did not explicitly report HF or AF. To our knowledge, no meta-analysis has been performed to evaluate the relationship between early menopause and HF and AF. Some observational studies have reported inconsistent conclusions. The Women's Health Study reported a lower relative risk of AF in women with a younger age at menopause after multivariable analysis [12]. This is in contrast to the findings of a study including 638,229 women aged over 40 years without prior AF, in which a shorter duration of endogenous estrogen exposure was significantly linked to an increased risk for AF after 6.3 years of follow-up on average [13]. Women spend about a third of their lives after menopause, and much longer if they experience early menopause. Under the circumstances, it is important to examine the relationship between early menopause and HF and AF [14].

The object of this study was to comprehensively review the population-based cohort studies that have been conducted to assess the association between early menopause and a higher risk of AF and HF, and to get preliminary quantitative data on this risk by meta-analysis.

2. Materials and methods

2.1. Study design

This systematic review and meta-analysis follows the guidelines of PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [15]. The protocol for this study was published in PROSPERO on December 14, 2022 (Registration number: CRD42022381342).

2.2. Search strategy

The PICO (Population, Intervention, Comparison, Outcome) components used as the inclusion criteria for the meta-analysis were as follows: (1) Population: postmenopausal women; (2) Intervention group: women with early menopause, either PM or EM; (3) Comparison group: postmenopausal women, with age at menopause as reference category; (4) Outcome: heart failure or atrial fibrillation. To identify eligible studies, a systematic search was conducted in three databases, Embase, PubMed, and Web of Science, covering the period from the establishment of the database until April 1, 2023, and using the following terms: (1) terms related to early menopause ("Menopause" [Mesh], "Primary Ovarian Insufficiency" [Mesh], "Early Menopause", "Age at Menopause", "Premature Menopause"); (2) terms related to heart failure and atrial fibrillation ("Atrial Fibrillation" [Mesh], "Heart Failure" [Mesh], "Auricular Fibrillation", "Persistent Atrial Fibrillations", "Cardiac Failure", "Heart Decompensation"). Detailed search strategies for the three databases are listed in the Supplementary Table 1.

2.3. Selection criteria

Articles were considered if they fulfilled the following criteria. (1) Cohort studies; (2) Exposure variables included women with PM or EM; (3) The primary outcome of interest was HF or AF; (4) The literature provided an effect size and 95 % confidence interval (CI) for the primary outcomes associated with exposure variables. The following studies were excluded: (1) Different study types (e.g., review, laboratory study, or intervention study); (2) had no control group (without normal age menopausal women); (3) the follow-up period was <3 years; (4) the sample size was <10. Moreover, if more than one article reported the same study, we included only the most detailed one.

2.4. Data extraction

Data extraction was conducted by two independent investigators

(XSJ and WYC), and any disputes were settled by consensus among the reviewers who were not involved in the initial search. The following information was extracted: region of studies; name of cohorts; time of article publication; first author name; sample of participants; number of cardiovascular events (HF and AF); menopause category; age at enrollment; follow-up duration; exposure assessment; primary outcome; adjusted factors (smoking and HRT/OC use); multiple adjusted hazard ratios (HR) and corresponding 95 % CI.

2.5. Risk of bias and study quality assessment

The Newcastle–Ottawa scale (NOS) was used to assess the quality of each eligible study, with a total score of 9 points. A score of ≥ 7 , 4–6 and ≤ 3 was considered high, moderate, and low quality studies, respectively [16]. Each of the included studies scored >7 points. Detailed results can be obtained in the Supplementary Table 2.

2.6. Statistical analysis

Random-effects meta-analysis was used to combine the data of each included study due to the clinical and methodological heterogeneity among studies. Heterogeneity was determined using the Cochrane chi-square test, and the I^2 statistic was employed to assess the degree of heterogeneity. The values of I^2 were <40 %, 40–60 %, and >60 %, corresponding to no, slight, and considerable degrees of heterogeneity, respectively [17]. Because the reference category for age at menopause varied between studies, we reorganized the initial reference categories into a reduced category. For example, the categories of age at menopause: <40 years, 40–44 years were combined to <45 years [18]. The result was expressed as a HR with corresponding 95 % CI. We performed subgroup analyses according to age of menopause (<40 , <45 , and 45–49), region of study (Western or Asian countries), type of menopause (natural or surgical), and adjustment for confounding factors. Egger's test was used to assess publication bias. All analyses were performed with the use of Stata software (version 15.1, Stata Corp. LP, College Station, Texas, USA).

3. Results

3.1. Identification of relevant articles

The initial search provided 1478 articles, including 259 duplicates. After scanning the titles and abstracts, 1133 were excluded because they did not fulfill the requirements for inclusion. The remaining 77 papers were eligible for full-text screening, and 68 papers were excluded for the following reasons: (1) Other languages ($n = 1$); (2) Similar studies ($n = 3$); (3) Conference abstract ($n = 6$); (4) Other outcomes ($n = 11$); (5) Not cohort study ($n = 20$); (6) Different exposure variables ($n = 27$). Finally, a total of 9 articles were included in this study (Fig. 1).

3.2. Study characteristics

After the literature screening, a total of 9 cohort studies, including 6,255,783 postmenopausal women, were eligible for inclusion in this analysis [5,6,12,13,19–23]. Five studies [12,19,21–23] were performed in the US, two [6,13] in Korea, one [5] in the UK, and one [20] in Sweden. The number of articles with HF or AF as the primary outcome was six and five, respectively. Seven articles [5,6,12,13,19,22,23] included smoking as a confounder, one article [21] had unknown smoking status, and one article [20] analyzed women who smoked and never smoked separately. Use of hormonal therapy was considered as a confounding factor in five articles [5,6,19,22,23], unknown in one article [21], and women treated with and without hormone therapy were analyzed separately in three articles [12,13,20]. One article [19] pooled data from three studies, including Atherosclerosis Risk In Communities study (ARIC), Multi-Ethnic Study of Atherosclerosis (MESA),

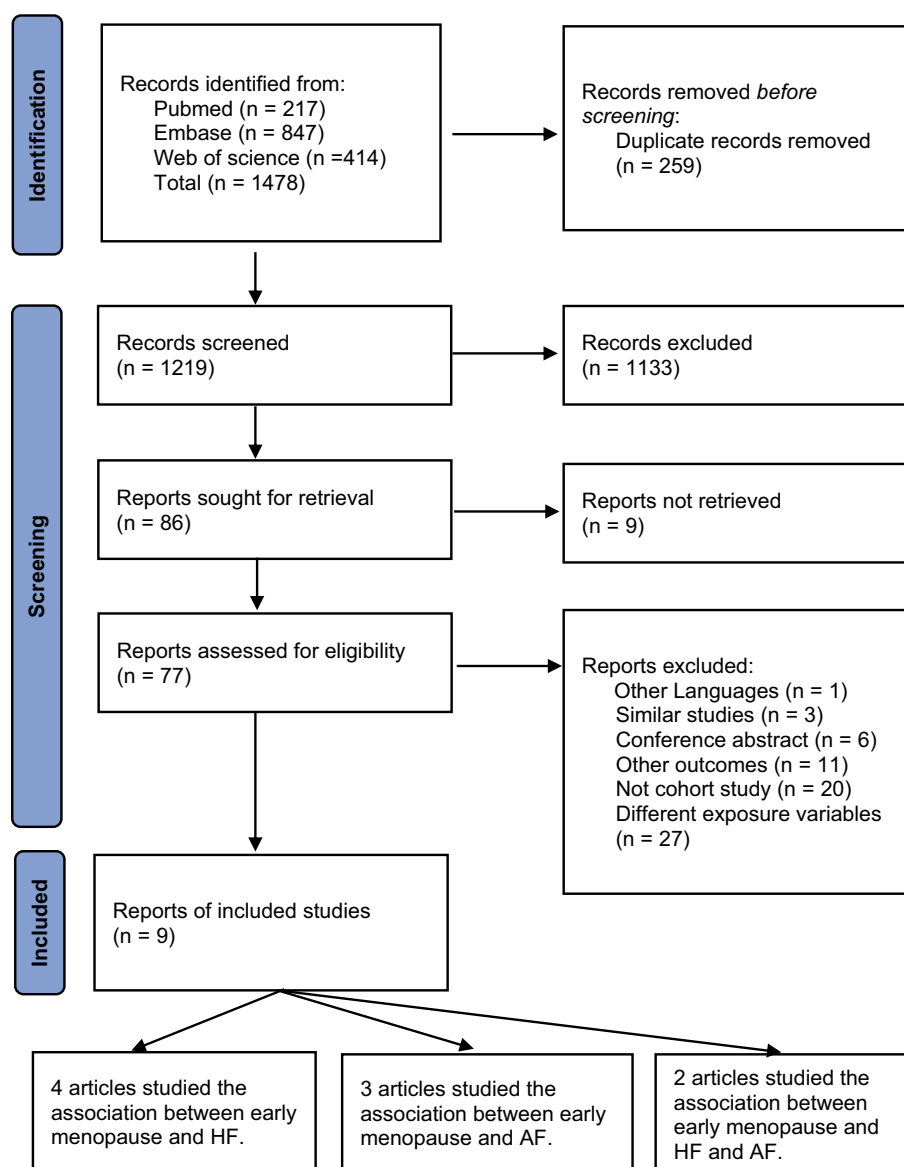


Fig. 1. Flowchart of included studies.

and the Jackson Heart Study (JHS). ARIC and MESA have been included in our analysis, so we only analyzed the data from JHS. The baseline characteristics of the included studies, such as first author name, name of cohort, number of participants, and primary outcome were listed in the Table 1.

3.3. Meta-analysis results

3.3.1. Heart failure

The association between PM and HF was reported in two articles and EM in six articles. Women with PM or EM had a higher risk of HF compared with women with menopausal age > 45 years (HR: 1.39, 95 % CI: 1.31–1.47, I^2 : 0 %; HR: 1.23, 95 % CI: 1.10–1.37, I^2 : 46.1 %, respectively) (Fig. 2). No significant publication bias was detected with the Egger's test ($P = 0.950$ for EM).

3.3.2. Atrial fibrillation

The association between PM and AF was reported in two articles and EM in five articles. Women with PM or EM had a higher risk of AF compared with women with menopausal age > 45 years (HR: 1.15, 95 % CI: 1.01–1.31, I^2 : 38.3 %; HR: 1.08, 95 % CI: 1.04–1.13, I^2 : 56.7 %,

respectively) (Fig. 3). No significant publication bias was detected with the Egger's test ($P = 0.819$ for EM).

3.4. Subgroup analysis

Subgroup analysis was performed according to age of menopause (<40, <45, and 45–49), study region (Western or Asian country), forms of menopause (surgical or natural), and adjustment for HRT or smoking (yes or no). Compared with normal-age menopausal women, women with menopausal age between 45 and 49 years had a higher risk of developing HF but not AF (HR: 1.11, 95 % CI: 1.09–1.14, HR: 1.00, 95 % CI: 0.88–1.13, respectively). Due to the limited number of studies included, we do not have sufficient evidence to indicate whether the type of menopause and different study regions significantly affect the relationship between early menopause and an increased risk of HF or AF. Subgroup analyses also showed that adjustment for confounding factors (HRT/OC and smoking) did not significantly affect the relationship between early menopause and HF or AF. Detailed results of the subgroup analyses are available in Fig. 4 and Fig. 5.

Table 1
Characteristics of the included cohort studies.

Author/year	Region	Cohort	No. of events HF/AF	Sample	Menopause category	Age at enrollment	Follow-up duration	Exposure assessment	Primary outcome	Adjusted factors		NOS scores
										Smoking	HRT/OC	
Appiah et al. 2016	USA	Atherosclerosis Risk In Communities study (ARIC)	965/-	5629	<45, ≥45r	56 (mean)	21.4 (median)	Questionnaire	HF	Confounder	Confounder	8
Ebong et al. 2014	USA	Multi-Ethnic Study of Atherosclerosis (MESA)	71/-	2947	<45, >45r	45-84	8.5 (median)	Questionnaire	HF	Confounder	Confounder	7
Honigberg et al. 2019	UK	UK Biobank cohort	772/2210	144,260	<40, ≥40r and <45, ≥45r	59.9 ± 5.4	7 (median)	Questionnaire	HF, AF	Confounder	Confounder	8
Magnani et al. 2012	USA	The Framingham Heart Study	-/273	1809	<45, 45-53r, >53	71.4 ± 7.6	10 or until incident AF	Questionnaire	AF	Not available	Not available	6
Rahman et al.2014	Sweden	Swedish Mammography Cohort	2532/-	22,256	40-45, 46-49, 50-54r, ≥55	49-83	13 (mean)	Questionnaire	HF	Separate analysis	Separate analysis	8
Shin et al. 2022	Korea	Korean National Health Insurance Service (NHIS)	42,699/ 44,834	1,401,175	<40, 40-44, 45-49, ≥50r	47-73	9.1 (mean)	Questionnaire	HF, AF	Confounder	Confounder	8
Wong et al. 2017	USA	The Women's Health Study (WHS)	-/1350	30,034	<45, 45-49, 50-54, >54r	53 (median)	20.5 (median)	Questionnaire	AF	Confounder	Separate analysis	6
Yang et al. 2019	Korea	Korean National Health Insurance Service (NHIS)	-/114,373	4,638,299	<45, ≥45r	52.5-75.7	6.3 (mean)	Questionnaire	AF	Confounder	Separate analysis	8
Yoshida et al. 2021 ^a	USA	Three pooled cohorts	1412/-	9374	<45, ≥45r	57.7 ± 8.6	15 (median)	Questionnaire	HF	Confounder	Confounder	8

HRT: Hormone replacement therapy; OC: oral contraceptive; NOS: Newcastle-Ottawa Scale; HF: Heart Failure; AF: Atrial Fibrillation; r: reference.

^a This article pooled data from three studies, including ARIC, MESA, and the Jackson Heart Study (JHS). ARIC and MESA have been included in our analysis, so we only analyzed the data from JHS. Note: Study characteristics are the whole study, not JHS alone.

4. Discussion

To the best of our knowledge, the current study, which included 9 cohorts of 6,255,783 postmenopausal women, is the first systematic review and meta-analysis to assess the relationship between early menopause and the risk of AF and HF. Our results suggest that women with PM or EM have an increased risk of HF and AF than those with menopausal age > 45 years.

The published evidence on the relationship between early menopause and a higher risk of AF is inconsistent. The Women's Health Study (WHS), examining a total of 30,034 females without CVD at baseline, reported an inverse association (HR: 0.82, 95 % CI: 0.67-1.02) between early menopause and the risk of AF after a median follow-up of 20.5 years, which is not consistent with the conclusions of our study [12]. However, the finding from WHS may be limited for several reasons. First, the population included in the study was composed entirely of female health care providers, who had more opportunities to intervene in the disease process and thus may have had a lower incidence of AF. Second, confirmation of AF required evidence of electrocardiography or outpatient or inpatient medical records documenting a history of AF, but medical records were obtained for only 80 % of self-reports. In the remaining 20 %, some women may have been diagnosed with AF but were not included in the final analysis, and it can be inferred that the detection rate of AF will be lower than expected. Although it is generally recognized that early menopause is linked to a higher risk of CVD, its relationship with HF has not been examined in detail in previous studies. In a cross-sectional study including 71 premenopausal and 72 age-matched postmenopausal women found that postmenopausal hormonal changes may impair left ventricular systolic and diastolic function and right ventricular diastolic function after echocardiographic measurements of cardiac parameters [24]. Similarly, another study including 34 middle-aged women, reported a significant reduction in left ventricular mechanics (torsion, twist velocity, and apical circumferential strain rate) in postmenopausal women compared with premenopausal women. [25] Emerging large sample epidemiological evidence suggests a positive correlation between early menopause and HF events. Shin's study, which examined 140,175 postmenopausal women, found that women with a history of early menopause had an increased risk (HR: 1.30, 95 % CI: 1.16-1.60) of HF after 9.1 years of follow-up on average, compared with women without a history of early menopause [6].

The relationship between early menopause and an increased risk of HF and AF may be explained by the following mechanisms. Firstly, it has been established that endogenous estrogen has a protective effect on the cardiovascular system. Estrogen can increase vasodilation and inhibit the vascular response to damage and the expression of inflammatory factors [26]. Besides, early estrogen loss may promote the development of arterial stiffness and increase the risk of CVD [27]. Second, shorter estrogen exposure is often linked to a higher risk of hypertension and diabetes, which are important and independent risk factors for the cardiovascular system [28]. Meanwhile, menopause may also change the body's visceral fat content, reduce insulin sensitivity, and increase the risk of metabolic syndrome [29]. A recent meta-analysis, including 21 observational studies, reported significant increases in low-density lipoprotein, triglycerides, total cholesterol, and fasting blood glucose in POI patients compared with control women, and insulin levels were slightly higher in POI patients [30]. In addition, endogenous estrogen depletion is often accompanied by hyperandrogenism and low serum hormone-binding globulin, which are linked to CVD [31].

The subgroup analysis of this study shows that, compared with EM, PM has a stronger association with the high risk of HF and AF, which can be explained by the exposure time of estrogen. Different types of menopause may affect the incidence of HF and AF. In women with bilateral oophorectomy, hormone levels fall abruptly, leading to acute hypoestrogenemia and hypoandrogenemia. In naturally menopausal women, hormone levels usually decline slowly, and low concentrations of estrogen can still be measured in the periphery for many years after

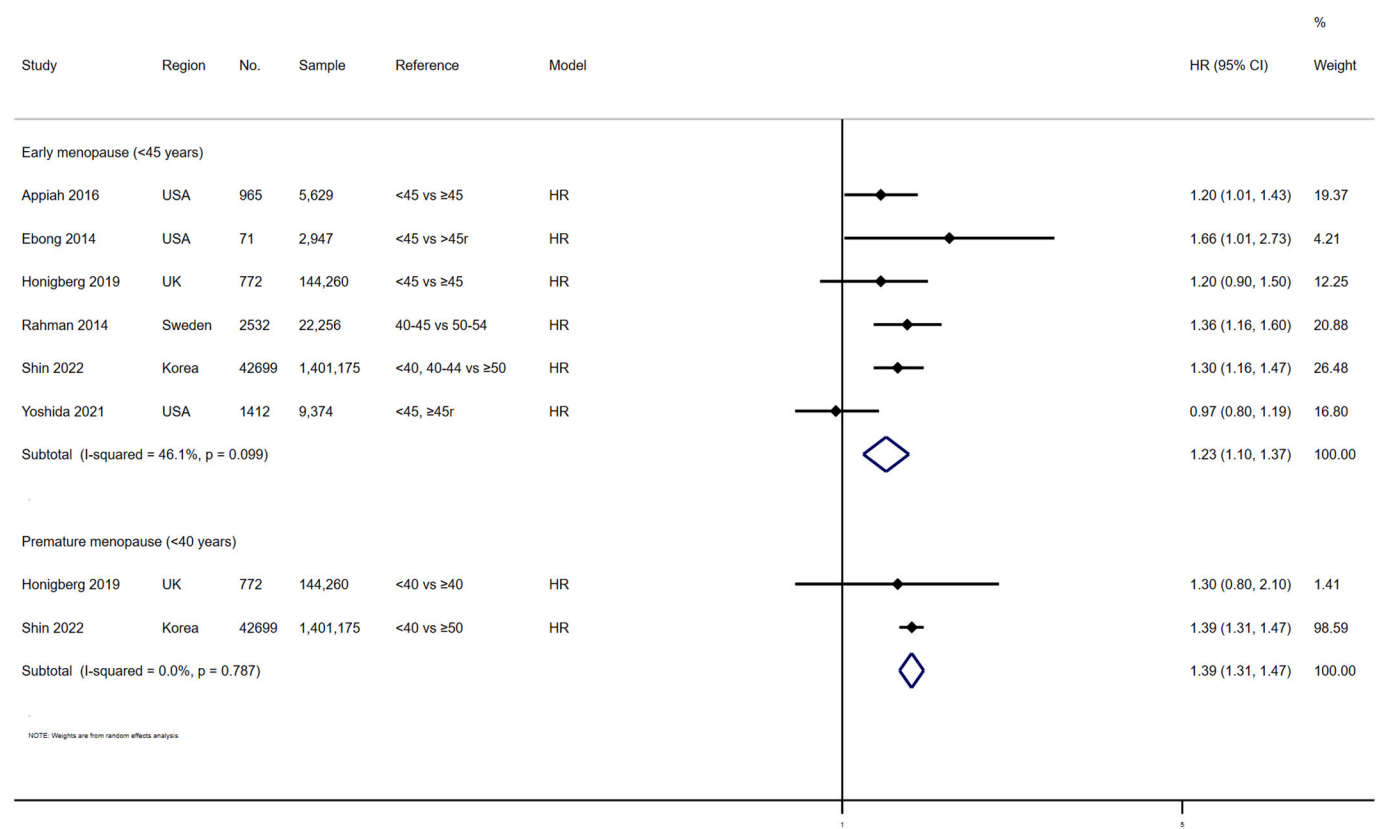


Fig. 2. Forest plot presents the association between PM or EM and the risk of HF in cohort study.

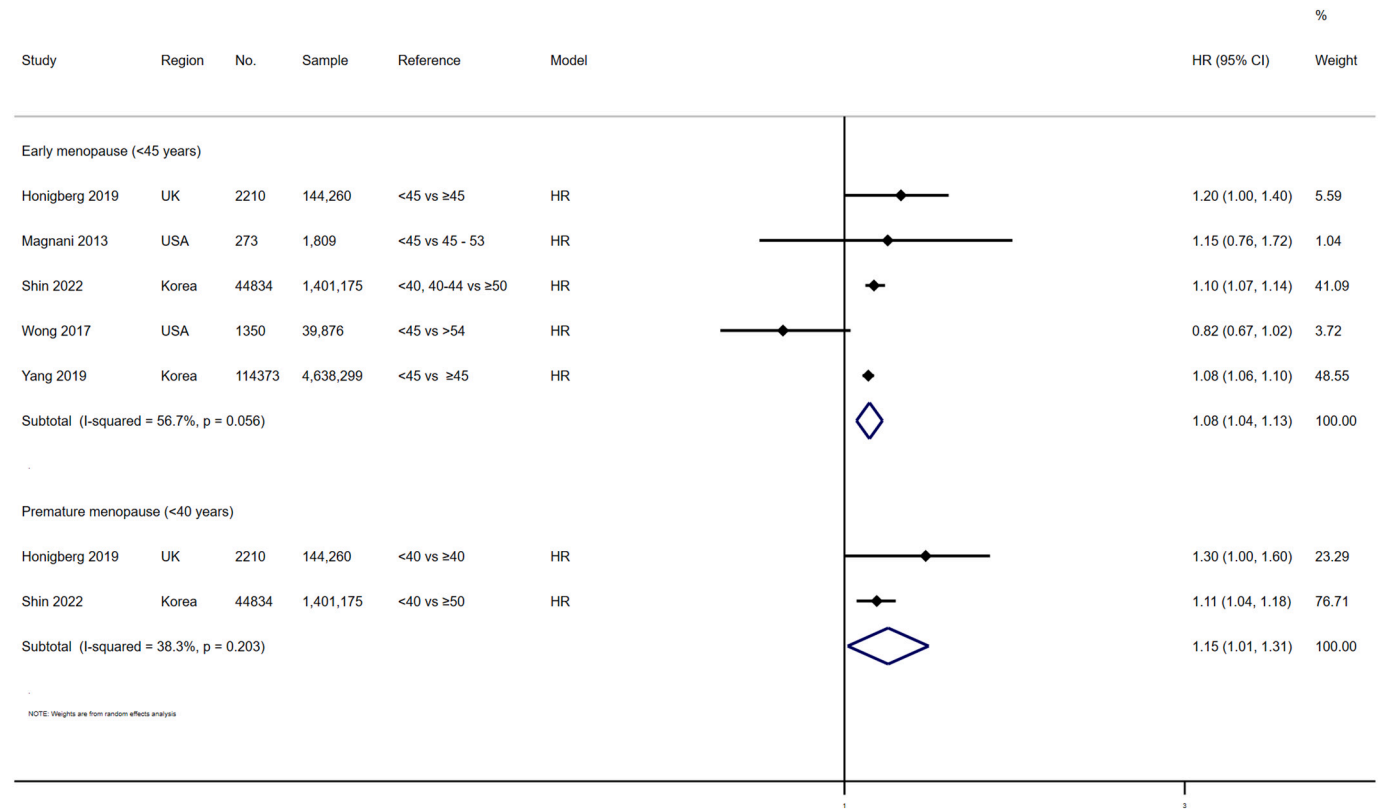


Fig. 3. Forest plot presents the association between PM or EM and the risk of AF in cohort study.

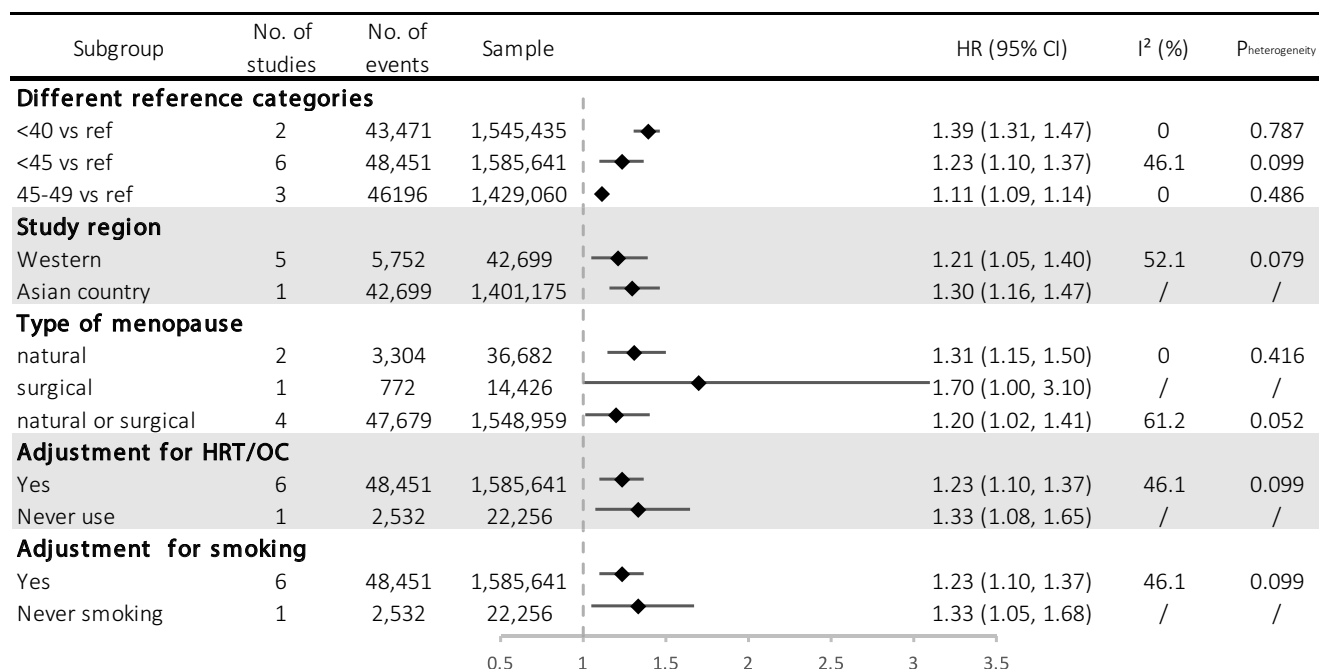


Fig. 4. Subgroup analysis between EM and risk of HF.

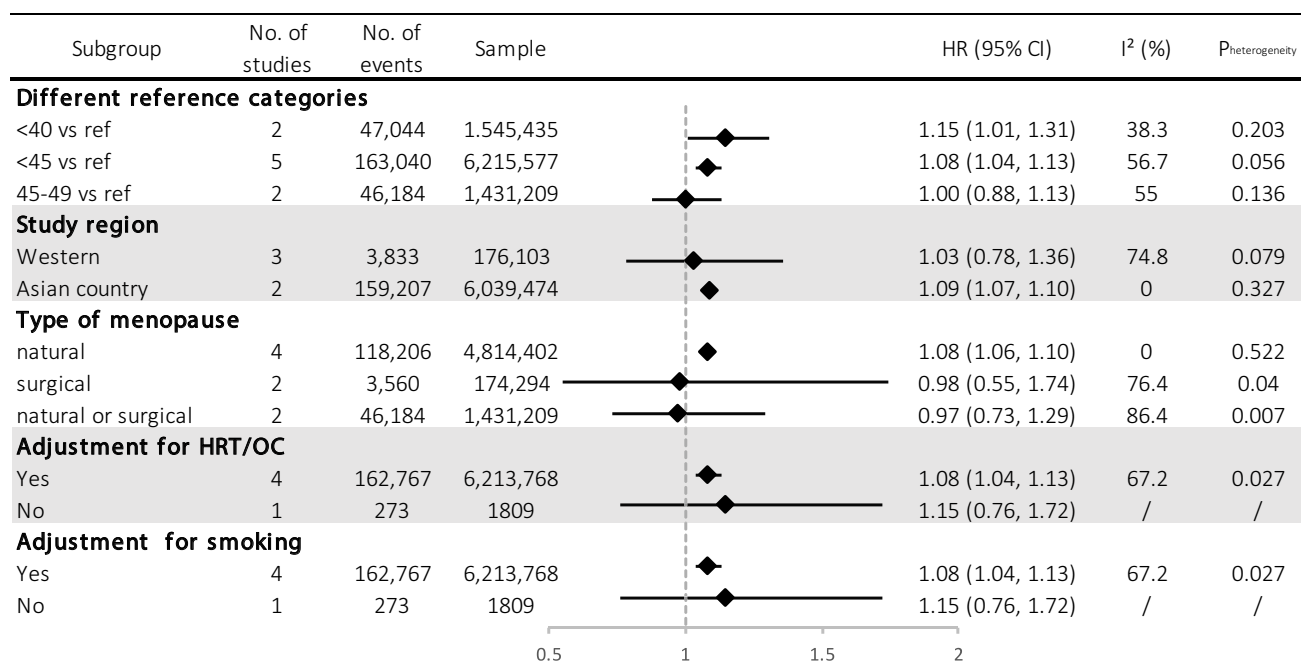


Fig. 5. Subgroup analysis between EM and risk of AF.

menopause [32]. This conclusion implies that different types of menopause may affect postmenopausal hormone levels, which are associated with cardiovascular events. However, conclusions regarding the effect of different types of menopause on cardiovascular risk are inconsistent. One study involving 10 international cohorts showed that surgical menopause was associated with a 20 % increased risk of CVD compared with natural menopause (HR: 1.22, 95 % CI: 1.16–1.28) [33]. Another study that specifically addressed the association of menopause mode with CVD showed that the difference between natural and surgical menopause and cardiovascular risk did not exist. Both studies suggest

that the cardiovascular risk profile before surgery may already determine the future cardiovascular risk of women undergoing oophorectomy, with little to do with the surgery itself. Their conflicting findings may be related to different baseline characteristics at enrollment. As Honiqberg explains, the conclusion that the type of menopause was little different from cardiovascular risk may have been due to similar baseline cardiovascular characteristics at enrollment between the natural and surgical menopause groups. In our meta-analysis, only a limited number of studies [5] specifically reported the effects of natural and surgical menopause on HF and AF, which may affect the stability of the results,

so our conclusions regarding subgroup analyses of types of menopause must be interpreted with caution. Different regions may influence the association between early menopause and HF and AF. However, most of the participants were Western women, and there was insufficient evidence to determine whether demographic and geographic characteristics contributed to the association.

The main strength of this study is the high quality of the included articles, each of which scored above 7 points according to the NOS. In addition, the sample size included in this analysis is large, so the reliability of the conclusion is higher. Our study also has some limitations. First, the participants including in the analysis only included women who were willing to accept questionnaires and agreed to long-term follow-up, which may cause selection bias [12]. Second, the age at menopause was self-reported in all included studies, which may have produced recall bias. However, published study suggests that self-reported age at menopause has a reasonably high level of validity and repeatability, with the majority of women describing their age at menopause accurately [34]. Besides, standard questionnaires and detailed questions were applied in each article, so we consider that the heterogeneity of age at amenorrhea across studies is relatively small. Third, several studies [6,13] included participants from the health insurance service cohort, which is a fee-for-service medical service that is reimbursed based on diagnosis codes and medical records. Sometimes patients choose to exaggerate their condition in order to obtain a higher reimbursement amount, resulting in incorrect diagnosis by the physician, which may affect the incidence of HF and AF. Fourth, potential confounders may affect the reliability of the outcomes. Confounding factors such as form of menopause (natural or surgical), HRT, and smoking may affect the relationship between early menopause and a higher risk of HF or AF. However, because the subgroup analysis included only a few articles, the precise contribution of confounding factors to the outcomes cannot be determined. Finally, according to the latest European Society of Cardiology guidelines, HF consists of three subtypes including reduced ejection fraction, and AF can also be divided into different types such as persistent and paroxysmal [35,36]. Our study does not provide additional evidence for the risk associated with early menopause and different subtypes of HF or AF. Therefore, the association between early menopause and different subtypes of HF and AF could be investigated in the future.

The main public health values of this study are as follows: First, it is well known that diseases such as gestational diabetes mellitus and polycystic ovarian syndrome are unique cardiovascular risk factors for women [37]. According to the American Heart Association and the American College of Cardiology, menopause before the age of 40 is a factor for enhanced cardiovascular risk [38]. Therefore, in addition to traditional cardiovascular risk factors, a history of gynecological diseases such as early menopause should also be considered when assessing a woman's future risk of HF or AF. Second, early detection of women with early menopause who may be more susceptible to HF and AF, offers a time window for early intervention. Timely and effective lifestyle interventions may prevent early menopausal women from developing HF or AF after menopause and thereby reduce the health care budget. Third, despite substantial evidence suggesting that endogenous estrogen has positive effects on the cardiovascular system, the treatment effect of exogenous estrogen therapy after menopause is still controversial to some extent. In the Women's Health Initiative study, HRT was not cardio-protective, and similar findings were shown in the secondary prevention trial of the Heart and Estrogen/Progestin Replacement Study [39,40]. Besides, these studies have reported that HRT increases the incidence of breast cancer and CVD [41]. In contrast, other studies have reported that it reduces CVD mortality and slows the evolution of carotid intima-media thickening [6]. These contradictory conclusions provide a preliminary foundation for further investigation of the complex mechanism of exogenous estrogen in the occurrence and development of CVD.

5. Conclusion

In conclusion, this first systematic review and meta-analysis reports that women with PM or EM have a higher risk of HF and AF compared with women with menopause age > 45 years. This finding may facilitate the early detection of women at increased risk of HF and AF and the timely implementation of lifestyle interventions to reduce the incidence of HF and AF. However, the exact mechanism between early menopause and a higher risk of HF and AF needs to be elucidated by larger sample sizes and more well-designed prospective studies.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.maturitas.2023.107784>.

Contributors

Jiajun Liu designed the study, analyzed the feasibility of the study, and wrote the manuscript.

Xueshan Jin searched the literature, extracted the data, assessed the quality of literature, and conducted the statistical analysis.

Wanying Chen searched the literature, extracted the data, assessed the quality of literature, and conducted the statistical analysis.

Lan Wang searched the literature, extracted the data, assessed the quality of literature, and conducted the statistical analysis.

Ziyi Feng searched the literature, extracted the data, assessed the quality of literature, and conducted the statistical analysis.

Jieming Huang reviewed the manuscript, provided critical scientific input, and checked the whole process of the study.

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Ethical approval

This manuscript has not been published and is not under consideration for publication elsewhere. All the authors have approved the manuscript and agree with this submission.

Provenance and peer review

This article was not commissioned and was externally peer reviewed.

Declaration of competing interest

All the authors declare that there are no conflicts of interest that could be seen as compromising the objectivity of the study.

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